M A C H I N E L E A R N I N G I

Master in Business Data Science — Group Assignment

Classification Models for Rental Prices in Madrid

Segmentation, Association Analysis & Logit Regression

Group 3

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1. Executive Summary

This report focuses on a two-phase Machine Learning study in 2'085 rental listings in Madrid, with data extracted from Idealista.com. This report will support a real estate agency's portfolio and pricing decisions.
Phase 1 applies unsupervised clustering to break up the actual market structure. Phase 2 then builds a Logistic Regression classifier to identify properties that break the €1'800 rent per month threshold. This barrier separates the agency's standard from its premium portfolio. Both phases point towards two dominant characteristics: central location and property size.

Key Findings

- **Market segments:** K-Means (k= 5) produced three different groups: Entry-Level Interior (251 properties, 12%), High-Rise Exterior (238 properties, 11.4%), Grand Estate (163 properties, 7.8%), Standard Exterior Living (1'210 properties, 58%), and Urban Premium (223 properties, 10.7%).
- **The Grand Estate is the smallest in terms of volume but the most commercially valuable,** with a median rent of €4'500 per month.
- **Location is the most dominant predictor.** A property located in a central district is approx. 7.83x more likely to achieve high rent with about €700 more rent per month. This poses by far the largest single effect across both performed frameworks.
- **Size grows steadily.** The odds of a higher rental price grow with each additional square meter (sqm) by approx. 1.05x (~5%). In the Linear Regression, this adds approx. €10 per month for every sqm increase.
- **Model performance:** The model performs strongly with an AUC score of approx. 0.94, which confirms the discriminatory power with a small train-test gap, which rules out overfitting. Additionally, the Linear Regression model explains approx. 78% (R^2≈0.78) of the variation in the rent prices from the property characteristics alone.

2. Data Audit & Segmentation

2.1 Data Preparation

The raw dataset from Idealista.com contained 2'100 property advertisements. After removing 15 duplicates or structurally invalid entries, the working dataset consists of 2'085 properties with 14 variables. The variables contained physical attributes (bedrooms, etc.), amenities (elevator, etc.), and location identifiers (area, etc.).
Four features were specifically engineered: Is_Special, Is_Studio, Is_Central, and Price_per_sqp. Is_Special consolidates the four non-standard property types of penthouse, cottage, duplex, and semi-detached, covering 321 (15.4%) properties. Is_Studio was extracted through pattern-matching the free-text address field, leading to 92 (4.4%) studios being consolidated. Is_Central classified 868 (41.6%) properties as being located in Madrid's inner ring, based on the district-level premium threshold. Price_per_sqm was not used as clustering input as it was based in the rent and would introduce circular reasoning into the segmentation. It was purely used for profiling purposes.

2.2 Clustering Approach and Algorithm Selection

Clustering was performed on five property features: Sq. Mt, Bedrooms, Floor, Outer orientation, and Is_Special. In order to reflect structural property types and not just price levels, the features, Rent, price per sqm, and district were deliberately excluded. The single notable correlation was found between Sq. Mt and Bedrooms (r = 0.75). This, however, was to be expected and accepted due to the different dimensions being captured respectively. Additionally, all features were standardized (mean = 0, std = 1) before the fitting.
Five algorithms were evaluated. K-Means at k=5 produced a silhouette of 0.4453 with cluster sizes of 251, 238, 163, 1'210, and 223. K=4 produced a slightly higher silhouette = 0.4961.

K=5 cluster was chosen for two reasons. The first being that one cluster in k=4 covers 67.1% of the properties, while k=5 reduces that number to 58%. Second, being that k=5 catches the height premium, which separates Standard Exterior (median floor 2, €1'300) from High-Rise Exteriors (median floor 7, €1'600). Hierarchical clustering at k=5 produced an ARI of 0.6. DBSCAN found 3 clusters with 116 noise points, identifying luxury outliers with a median rent of €3'950 versus €1'395 for the main dataset. As this might be a useful insight overall, for a business segmentation, this is very much impractical. GMW and Spectral Clustering underperformed in interpretability and inter-algorithm agreement.

2.3 Segment Profiles

The clusters were profiled after the clustering using the outcome variables rent, price per sqm, and district. The segment labels are Entry-Level Interior (Cluster 0), High-Rise Exterior (Cluster 1), Grand Estate (Cluster 2), Standard Exterior Living (Cluster 3), and Urban Premium (Cluster 4).

	Entry-Level Interior (Cluster 0)	High-Rise Exterior (Cluster 1)	Grand Estate (Cluster 2)	Standard Exterior Living (Cluster 3)	Urban Premium (Cluster 4)
CLUSTER SIZES	251	238	163	1'210	223
SHARE OF MARKET	12%	11.4%	7.8%	58%	10.7%
MEDIAN RENT	€1'075	€1'600	€4'500	€1'300	€2'100
MEDIAN AREA	60 m2	100 m2	391 m2	88 m2	121 m2
KEY CHARACTERISTICS	Central, no views	High-floor, exterior views	Large special properties	Market volume, mid/low floor	Penthouse-heavy segment

Table 1. *K-Means cluster profiles (k=5).*

3. Supervised Model – Logistic Regression

3.1 Model Choice and Rationale

We chose the Logit Regression Model as the main supervised model rather than the Linear Regression Model, because the primary business goal is to classify properties with monthly rents over €1'800, which therefore requires a binary classifier instead of a continuous predictor.
By applying the Logit Regression Model, we get probabilities for each property. These outputs can be converted to a High-Low Rent classification at a given cutoff. Moreover, the Odds Ratios (Exp(b)) directly measures possible influence of every feature on High-Rent properties' likelihood, making it possible for a non-technical agency to take action immediately.

3.2 Modelling Pipeline

An 80/20 train-test split was applied in order to preserve the class balance across both of the sets. Additionally, a VIF threshold of 10 was introduced to guard against multicollinearity. All 8 of the features passed the with the highest score being obtained by Bedrooms (VIF=9.54).
The Stats model Logit was used in order to obtain the Wald test p-values, confidence intervals, and the Exp(b) Odds ratios. Afterwards, RFECV with a 5-fold cross-validation was used to confirm the optimal feature subset; it retained all 8 features, validating that none were redundant after the VIF check. The final evaluation was then based on the confusion matrix, ROC curve, and threshold sensitivity analysis.

3.3 Results – Odds ratio and model performance

With Is_Central, we found the strongest predictor. Being in a central location multiplies the odds of achieving a high monthly rental price by approx. 783. The predictor Sq. Mt adds approx. 5% to the odds with every additional sqm (~1.05x). Floor level is also a significant factor confirming that views from and lightning in the apartment command premium rental prices in the Madrid market. Bedrooms, Outer orientation, and elevators are not significant by themselves.

Variable	Exp(b)	p-value	Sig.	Business meaning
IS_CENTRAL	~8x	<0.001	***	Central location multiplies odds of heigh rental prices by ~8x
SQ.MT	~1.05x	<0.001	***	~5% increase in rental prices with every one unit increase in sqm
FLOOR	~1.13x	<0.001	***	Higher floors increase odds (premium views and better lightning)
IS_STUDIO	~0.07x	0.046	**	Studios reduce high rent by 93%
BEDROOMS/ OUTER ORIENTATION / ELEVATORS	-	>0.05	n.s.	Not individually significant

Table 2. *Logistic Regression Coefficient Summary (Odds Ratios and Significance)*

3.4 Connection to Association Analysis and Threshold Deployment

The results of the Logit Regression model confirm the initially found rules form the association analysis. Additionally, we find that Bedrooms alone is not independently significant.
The threshold of the model can be adjusted by the real estate agency. A lower threshold would result in more premium properties being detected (higher recall) while a higher threshold would reduce the false positives (higher precision).

3.5 Business Insight - Decision Matrix

This model is conservative in predicting high-rent properties. Take the confusion matrix at a cutoff 0.5 for an instance: with a specificity of 92.2% and a recall of 78.9%, the agency can correctly identify low-rent properties and capture high-rent ones at the same time. But also, it means that the model will miss roughly 21.1% real high-rent units (FN) while getting only 7.8% false alarms(FP) when predicting properties. This is exactly the reason why the prediction model is conservative, because it tends to underpredict a high-rent property instead of overpredicting it.
Besides, the cutoff of 0.5 reflects this trade-off. This cutoff level is balanced between accuracy, precision, and recall: if we take a cutoff of 0.4, the recall will be improved to 84%, but the precision will be lowered to approx.84% as well. Instead, if we set the cutoff to 0.6, the false positive rate will be reduced, but the agency will miss more high-rent properties. Thus, this decision matrix is the key validation for setting the prediction model.

	PREDICTED LOW	PREDICTED HIG
ACTUAL LOW (256)	TN = 236 (92.2%)	FP = 20 (7.8%)
ACTUAL HIGH (161)	FN = 34 (21.1%)	TP = 127 (78.9%)

Table 3. *Confusion Matrix(cutoff=0.5)*

This deliberately designed conservative model is especially meaningful in real-world business scenarios. In the Madrid rental market, compared to an overpriced property(FP), an underpriced property normally costs more, as it can lead to prolonged vacancy. The prediction model greatly decreases the risk of overpricing a property by taking accuracy(86.4%) as priority. As to 21.1% missed opportunities, the agency can still rely on experience and site visits to identify additional high-rent properties.

4. Conclusions

We can conclude that the rental market in Madrid consists of a stable five-tier structure, which is reliably identified by K-Means (k=5) and validated by Hierarchical clustering (ARI ≈ 0.60). Central location and property size are dominating as factors for the rental prices. The Linear Regression model explains approx. 73% of train variance (R^2=0.73) and 78% of test variance (R^2=0.78), confirming generalization without overfitting. The remaining 22% are attributed to variables outside of the dataset.

4.1 Recommendations

We suggest three agency suggestions based on this analysis. To automatically score premium listings, first use the Logit model as a Listing Screener during intake. Second, use the OLS model to apply an Under-pricing Flag to find properties that are substantially less expensive than the modeled fair value. Third, give priority to premium acquisition efforts by concentrating on the center districts' Grand Estate and Urban Premium segments, which make up 18.5% of all properties yet have a disproportionately high rental value.

4.2 Limitations

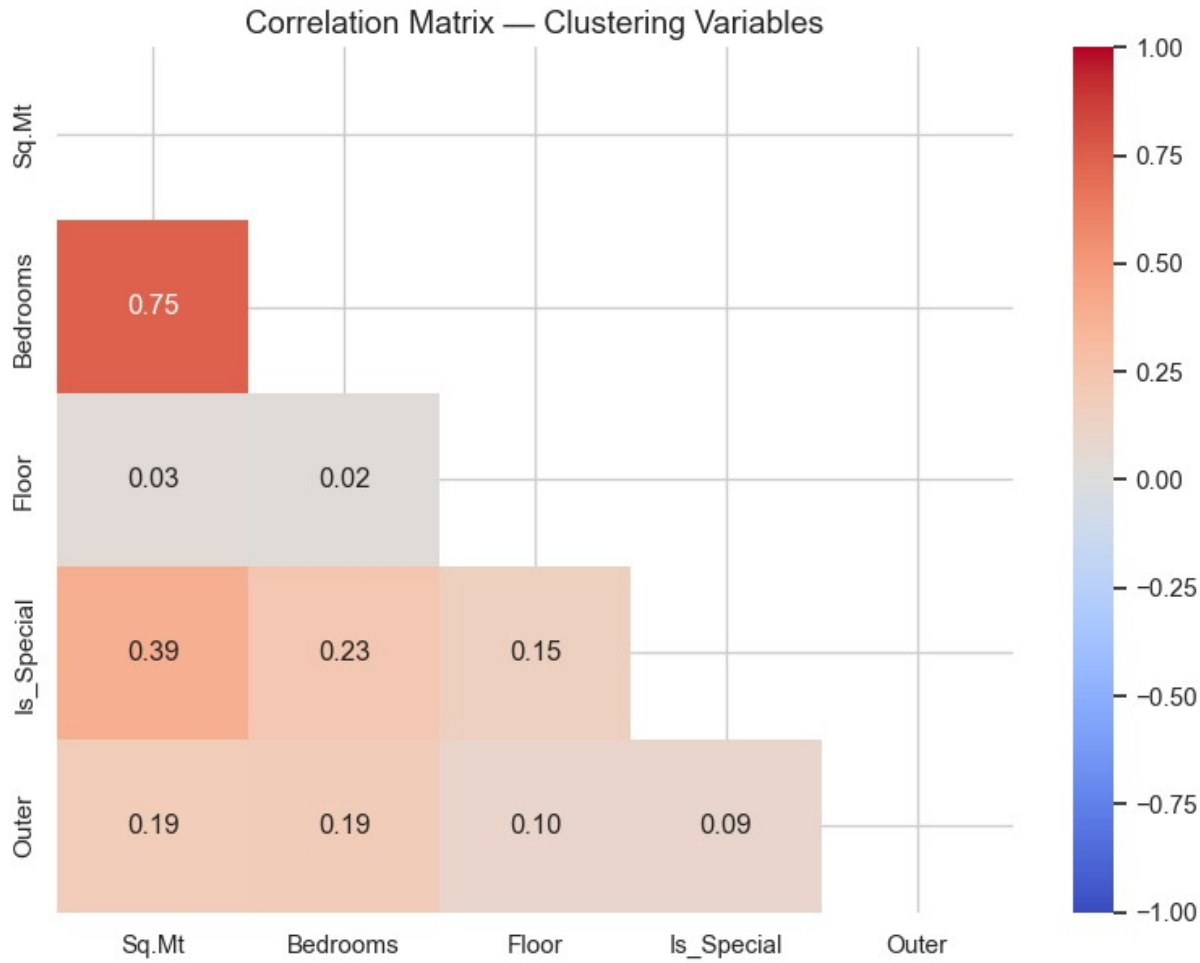
Several factors are not included in the original dataset (renovation quality, furnished status, etc.). These factors likely explain the missing 22% variance. Additionally, the coefficient values should be treated as discretional estimates and recalibrated periodically.

Technical Annex

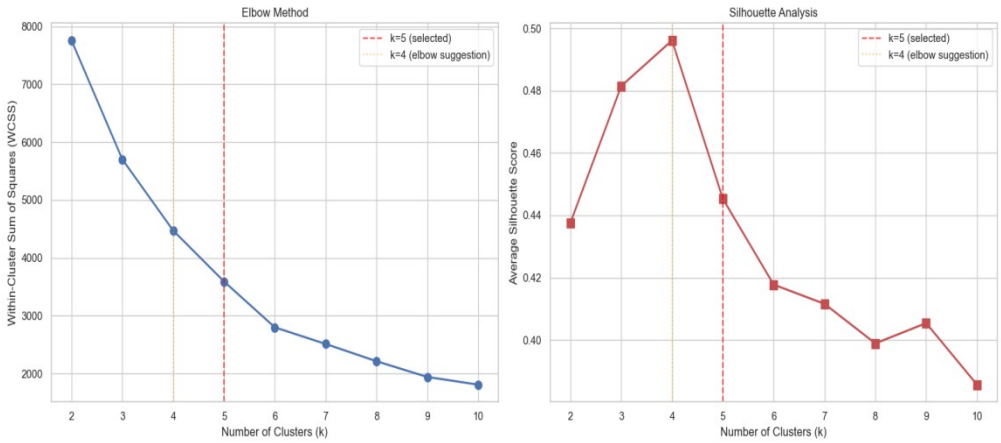
A.1 Cluster Validation — Segment Profiles (k=5 K-Means)

Cluster	Segment Label	n	Share (%)	Median Sq.Mt	Median Floor	Outer (%)	Is_Special (%)	Elevator (%)	Median Rent (€)	Mean Rent (€)	Median Price/m²
0	Entry-Level Interior	251	12.0%	60.0	3	0%	5%	75%	1075	1217	17.1
1	High-Rise Exterior	238	11.4%	100.0	7	99%	1%	99%	1600	1878	15.7
2	Grand Estate	163	7.8%	391.0	3	99%	51%	97%	4500	5090	11.7
3	Standard Exterior Living	1210	58.0%	88.0	2	100%	0%	88%	1300	1584	14.4
4	Urban Premium	223	10.7%	121.0	5	100%	100%	95%	2100	2372	15.6

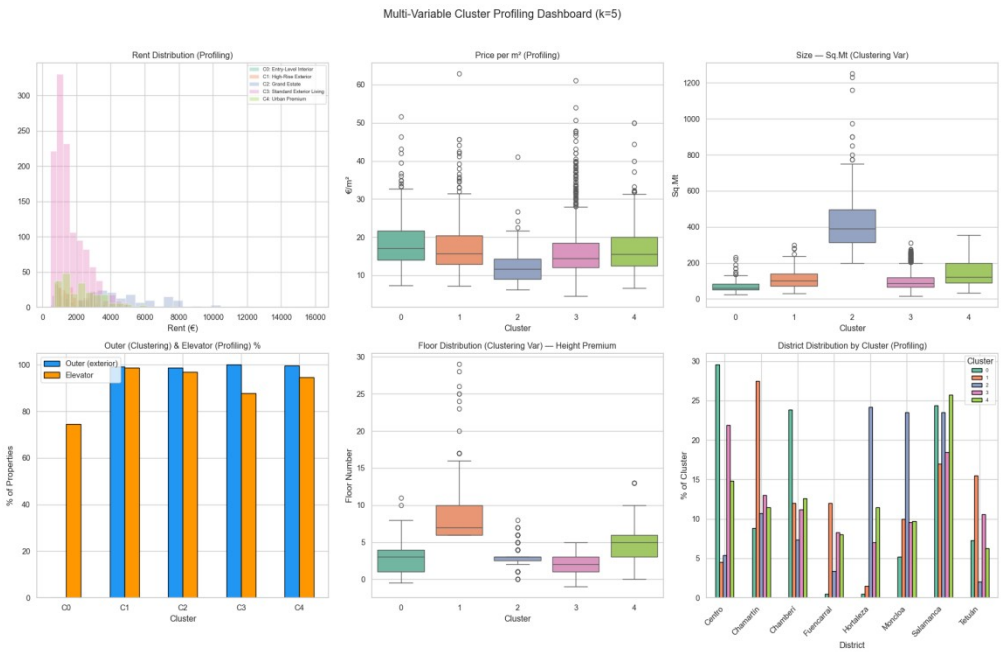
A.2 Correlation Matrix for Cluster Variables



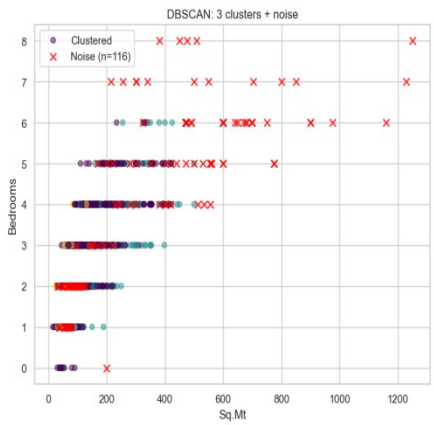
A.2 Elbow Method & Silhouette Analysis



A.3 Segmentation Clusters Profiles Dashboard



A.4 DBSCAN



Noise Points vs Clustered: Median Values

	Noise (outliers)	Clustered (normal)
Rent	3950.0	1395.0
Sq.Mt	332.5	90.0
Bedrooms	5.0	2.0
Price_per_sqm	14.2	15.0

A.5 VIF Analysis

VARIABLE	VIF
BEDROOMS	9.54
ELEVATOR	7.39
OUTER	6.60
SQ.MT	5.62
FLOOR	2.58
IS_CENTRAL	1.66
IS_SPECIAL	1.46
IS_STUDIO	1.07

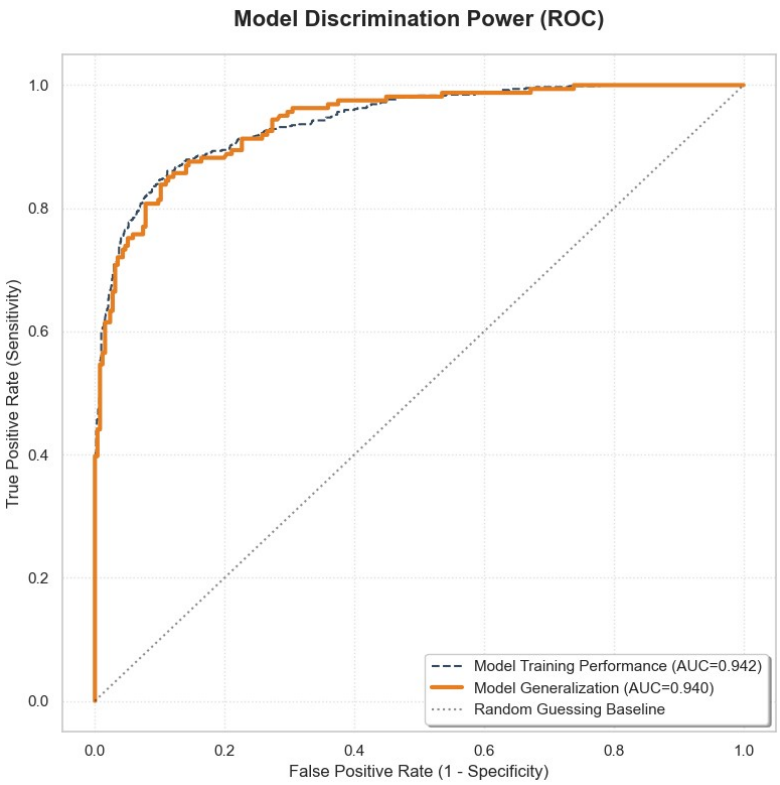
A.6 Odds Ratios & Confidence Intervals

	Beta	Exp(B)	p-value	95% CI Exp(B) Low	95% CI Exp(B) High	Significant
CONST	-7.233300	0.000700	0.000000	0.000300	0.001900	✓
SQ.MT	0.046800	1.047900	0.000000	1.041200	1.054500	✓
BEDROOMS	-0.138200	0.870900	0.250400	0.688100	1.102400	✗
FLOOR	0.124500	1.132600	0.000000	1.077400	1.190700	✓
OUTER	0.380400	1.462900	0.155400	0.865600	2.472300	✗
ELEVATOR	0.114900	1.121700	0.738700	0.571200	2.202700	✗
IS_SPECIAL	0.164200	1.178400	0.523900	0.711200	1.952600	✗
IS_CENTRAL	2.058600	7.834700	0.000000	5.448200	11.266600	✓
IS_STUDIO	-2.600400	0.074200	0.045700	0.005800	0.951500	✓

A.7 Logit Classification Performance on Held-Out Test Set

Metric	Value
AUC (TRAIN)	0.9418
AUC (TEST)	0.9403
AUC GAP	0.0015
ACCURACY	0.8705
PRECISION	0.8639
RECALL	0.7888
SPECIFICITY	0.9219
F1-SCORE	0.8247

A.8 ROC Curve



A.9 Top 10 Association Rules by Lift

ANTECEDENT	CONSEQUENT	SUPPORT	CONFIDENCE	LIFT
{CENTRAL, NO_OUTER}	**{Cluster_0, Elevator}**	**6.9%**	**83%**	**9.27**
{CLUSTER_0, ELEVATOR}	{CENTRAL, NO_OUTER}	6.9%	76%	9.27
{CENTRAL, CLUSTER_0}	**{Elevator, No_Outer}**	**6.9%**	**84%**	**9.23**
{ELEVATOR, NO_OUTER}	{CENTRAL, CLUSTER_0}	6.9%	76%	9.23
{OUTER, PENTHOUSE}	**{Cluster_4}**	**7.5%**	**96%**	**9.00**
{CLUSTER_4}	{OUTER, PENTHOUSE}	7.5%	70%	9.00
{CLUSTER_4}	**{Elevator, Outer, Penthouse}**	**7.1%**	**66%**	**8.98**
{ELEVATOR, OUTER, PENTHOUSE}	{CLUSTER_4}	7.1%	96%	8.98
{OUTER, PENTHOUSE}	**{Cluster_4, Elevator}**	**7.1%**	**91%**	**8.97**
{CLUSTER_4, ELEVATOR}	{OUTER, PENTHOUSE}	7.1%	70%	8.97

A.10 Interactive Rent Predictor Tool

As a business deliverable, the trained OLS and Logit coefficients were deployed into an interactive React application (Madrid_Rent_Predictor.jsx). The tool accepts property inputs (size, bedrooms, floor, exterior orientation, elevator, central zone, special type, studio) via sliders and toggles, and outputs in real time: (1) the predicted monthly rent from the Linear Regression model with 95% confidence interval, (2) the probability of being classified as High Rent (≥€1,800) from the Logit model, (3) the K-Means segment assignment, and (4) a feature impact waterfall showing each variable's euro contribution. This demonstrates how the analytical models can be operationalized for agency use.